

CLINICAL DECISION SUPPORT & MULTI-OMICS DOSSIER

Patient Submitter ID: TCGA-D1-A17D | Tumor: Skin Cutaneous Melanoma (SKCM) | Stage: Stage III

TCGA Project:	TCGA-SKCM	cBioPortal Study:	skcm_tcga_pan_can_atlas_2018
Gender / Age:	female / 52 yrs	Vital Status / OS:	Deceased (44.6 mo)
Hybrid ML Risk:	84.1% (CRITICAL RISK)	VQC / Classical:	80.5% / 86.5%

FOUNDATIONAL 6-DATABASE MULTI-OMICS MUTATION PROFILE (GDC, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl)

Gene	Locus (GRCh38)	Transcript	Protein Change	ClinVar Status	COSMIC ID	gnomAD Pop AF	VAF %
BRAF	chr7:7q34	ENST00000288602	p.V600E Kinase Hyperactivation	Pathogenic (FDA Approved Target) VCV000013961	COSV54483710 FATHMM: 0.99 Pathogenic	0.000011	44.2%
CDKN2A	chr9:9p21.3	ENST00000304494	p.R80* Nonsense Deletion	Pathogenic (p16INK4a Loss) VCV000009381	COSV52981042 FATHMM: 0.99 Pathogenic	0.000009	46.0%

QUANTUM VARIATIONAL ROTATION STATE ANGLES (θ_VQC / R_Y / CNOT)

Gene Symbol	Canonical Ensembl ID	VAF Hotspot Frequency	Variational Angle θ (Radians)	Qubit Hilbert Space Projection
BRAF	ENSG00000157764	44.2%	θ = 2.3020 rad (131.9°)	$ \psi\rangle = \cos(1.151) 0\rangle + \sin(1.151) 1\rangle$
CDKN2A	ENSG00000147889	46.0%	θ = 2.0550 rad (117.7°)	$ \psi\rangle = \cos(1.028) 0\rangle + \sin(1.028) 1\rangle$

PRECISION ONCOLOGY CLINICAL ACTION & PROTOCOL RECOMMENDATION

Recommended Therapeutic Pathway:

Combined BRAF+MEK inhibitor therapy (Dabrafenib + Trametinib) and anti-PD-1 immunotherapy evaluation.

Genomic Biomarker Summary: Identified somatic hotspot mutations in BRAF (p.V600E), CDKN2A (p.R80*). Confirmed pathological classification across ClinVar and COSMIC registries with extreme somatic rarity in gnomAD exomes.

NOTICE & RESEARCH DISCLAIMER: This report is generated by the BlazeFinix Quantum-Classical Multi-Omics Research Platform using open-access academic cancer genomics datasets (NCI GDC TCGA, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl). It is calibrated strictly for computational research, translational bioinformatics, and clinical decision support benchmarking, not for standalone autonomous medical diagnosis.